

Smooth a piecewise linear function

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1 The original function

The original function $f(x)$ is piecewise linear and takes the form:

$$f(x) = \begin{cases} 0 & x \leq x_1 \\ \left(\frac{c}{x_2 - x_1}\right)x - \frac{c}{x_2 - x_1}x_1 & x_1 < x \leq x_2 \\ c & x > x_2 \end{cases}$$

2 Smoothing scheme

The smoothed function $g(\cdot)$ with smoothing parameter s is

$$g(x; s) = \frac{c}{2} + \frac{1}{2} \left(\frac{c}{x_2 - x_1} \right) \left((x - x_1)^2 + \left(\frac{c}{x_2 - x_1} \right)^2 s^2 \right)^{\frac{1}{2}} \\ - \frac{1}{2} \left(\frac{c}{x_2 - x_1} \right) \left((x - x_2)^2 + \left(\frac{c}{x_2 - x_1} \right)^2 s^2 \right)^{\frac{1}{2}}$$